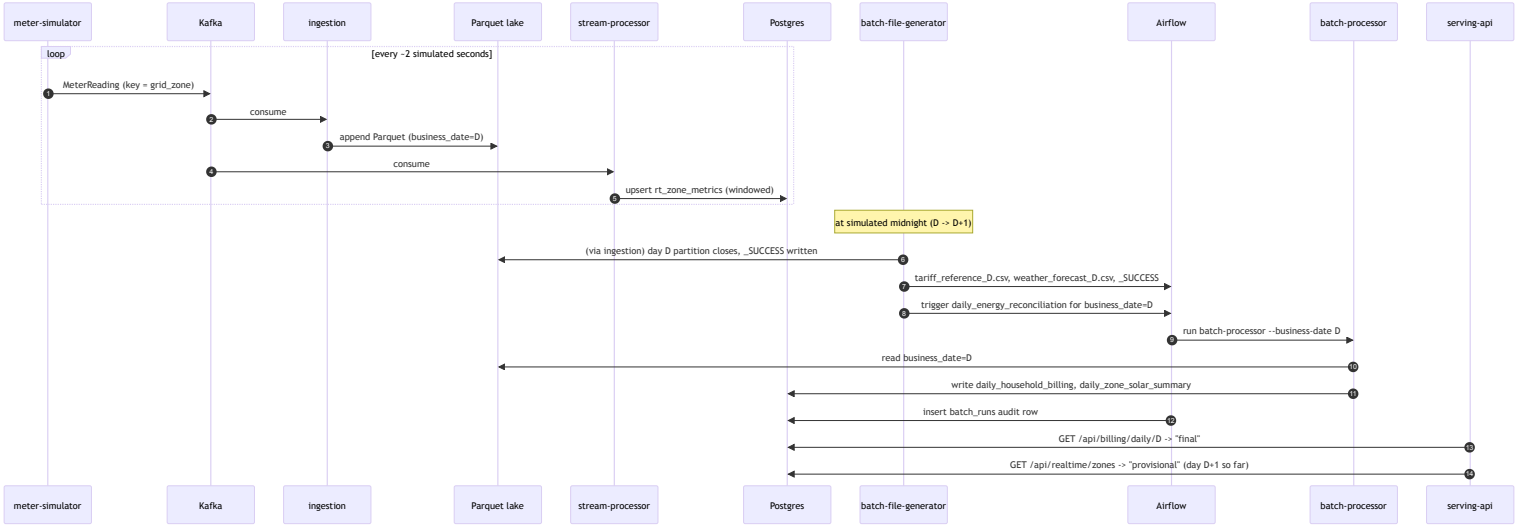
meter-simulator
Kafka
ingestion
Parquet lake
stream-processor
Postgres
batch-file-generator
Airflow
batch-processor
serving-api
loop
[every ~2 simulated seconds]
MeterReading (key = grid_zone)
consume
append Parquet (business_date=D)
consume
upsert rt_zone_metrics (windowed)
at simulated midnight (D -> D+1)
(via ingestion) day D partition closes, _SUCCESS written
tariff_reference_D.csv, weather_forecast_D.csv, _SUCCESS
trigger daily_energy_reconciliation for business_date=D
run batch-processor --business-date D
read business_date=D
write daily_household_billing, daily_zone_solar_summary
insert batch_runs audit row
GET /api/billing/daily/D -> "final"
GET /api/realtime/zones -> "provisional" (day D+1 so far)
1
2
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